Descriptor Similarity Does Not Predict Array Response: Chemical Warfare Agent Simulants Tested Against Seventeen Authentic Agents

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Abstract

Chemical warfare agent detection is developed and qualified on simulants, and we asked what measurement establishes that a simulant reproduces the agent it stands in for. A 29-dye colorimetric array was used to measure seventeen authentic agents, spanning nerve, blood, choking, vesicant and A-series compounds, at five concentrations in two optical channels, and the resulting response patterns were compared with the molecular-descriptor distances on which simulant selection in practice rests. Agreement was tested by exact Mantel permutation within each run, by exact enumeration of all 680 possible simulant designations in descriptor space, and against permutation nulls built for each comparison. The array separates compounds in all six run-channel cells, with p from 4.6×10^{-8} to 0.025, so the measurement is sound. Descriptor distance nonetheless fails to predict response distance: twelve exact Mantel tests and six

combinations of them leave nothing below $q = 0.05$. Dimethyl methylphosphonate, the standard G-agent simulant and the benchmark compound for decontamination efficiency, is anti-correlated with soman and cyclosarin in the ultraviolet channel, and its correlation with sarin falls from +0.77 at 500 ppm to +0.36 at 10 ppm, the end at which point detectors operate. Across the seventeen agents measured, exactly one designated simulant has ever been measured beside them.

Nearly every chemical warfare agent detector in the literature is developed, optimised and qualified on compounds that are not chemical warfare agents.[1,2] Dimethyl methylphosphonate stands in for the G-series nerve agents,[3] dipropylene glycol monomethyl ether for training detectors,[4] and methyl salicylate for vapour penetration through protective ensembles.[5,6] The substitution is unavoidable, because the agents are lethal at concentrations where handling them requires a licensed facility, and it is load-bearing: a sensor reported as selective for sarin has, in most cases, never been exposed to sarin.

What licenses the substitution is a claim of similarity. The compound chosen is said to be agent-like, and similarity is read from molecular descriptors: vapour pressure, dipole moment, frontier orbital energies, substructure and shape. The claim that matters for a detector is different. It is that the substitute produces the response the agent would produce in the same device. Those two claims are connected only if descriptor similarity transfers to response similarity, and that transfer has not been measured.

It has not been measured because measuring it requires the agents. Response data on authentic agents exist in few places, and the compounds used as simulants are rarely run alongside them. A recent survey of the field found no standard for chemical warfare agent simulants at all, and recorded that they have generally been selected on the basis of their historical use.[7] Our institute is an OPCW designated laboratory, which identifies and assays authentic agents under an accredited procedure, so the comparison can be made on the agents themselves.

This work reports that measurement. A 29-dye colorimetric array was used to record

the responses of seventeen authentic agents at five concentrations in two optical channels, and those responses were compared against the molecular-descriptor distances that simulant selection relies on. The array separates compounds in every run and channel, so the measurement is sound. Descriptor distance nonetheless carries no detectable information about response distance. Dimethyl methylphosphonate, the compound the field treats as the G-agent surrogate, is anti-correlated with two of the four G agents, and its resemblance to sarin weakens as the concentration falls towards the range where point detectors operate. Where the designation record and a regulatory toxicity model might have supplied the missing evidence, both are empty for the same compounds.

This is not a new sensing method, and we propose no replacement compound. It is a measurement of whether the substitutes the field already relies on carry the property they are relied upon for.

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Supporting Information Available

References

(1) Askim, J. R.; Mahmoudi, M.; Suslick, K. S. Optical sensor arrays for chemical sensing: the optoelectronic nose. *Chemical Society Reviews* **2013**, *42*, 8649.

(2) Rakow, N. A.; Suslick, K. S. A colorimetric sensor array for odour visualization. *Nature* **2000**, *406*, 710–713.

(3) National Institute for Occupational Safety and Health *Chemical Warfare Agent (CWA) Simulant Program for Respirator Materials Resistance Testing*; 2002; Dated 31 July 2002.

(4) Seitzinger, A. T.; Genovese, J. A. Chemical Agent Simulant Training Composition. 2006; https://patents.google.com/patent/US7129094B1/en, U.S. Patent US 7,129,094 B1, assigned to the United States Department of the Army. Filed 16 May 2003; granted 31 October 2006.

(5) National Research Council *Technical Assessment of the Man-in-Simulant Test Program*; The National Academies Press: Washington, DC, 1997.

(6) ASTM International Standard Test Method for Man-In-Simulant Test (MIST) for Protective Ensembles. ASTM F2588-12(2020), 2020; Designation F2588-12, reapproved 2020.

(7) Lawrence, D. S.; Van Houten, K. A. Simulants for Chemical and Explosive Threats. *Johns Hopkins APL Technical Digest* **2025**, *37*.